

## Problem 4.18

Let  $i$  be a simple interest rate and suppose that  $i_6 = 0.04$ . Calculate  $i$ .

### Solution

For simple interest,

$$a(t) = 1 + it.$$

The effective rate in period 6 is

$$i_6 = \frac{a(6) - a(5)}{a(5)}.$$

Since  $i_6 = 0.04$ ,

$$0.04 = \frac{a(6) - a(5)}{a(5)}.$$

Using  $a(t) = 1 + it$ ,

$$0.04 = \frac{(1 + 6i) - (1 + 5i)}{1 + 5i}.$$

So,

$$0.04 = \frac{i}{1 + 5i}.$$

Multiplying through by  $1 + 5i$  gives

$$0.04(1 + 5i) = i.$$

Hence,

$$0.04 + 0.20i = i.$$

Therefore,

$$0.04 = 0.80i,$$

and so

$$i = \frac{0.04}{0.80} = 0.05.$$

Thus,

$$\boxed{i = 0.05 = 5\%}.$$